

VIVEK PATEL

☎ 587-938-7059 ✉ patelvivek2372@gmail.com 🔗 linkedin.com/in/vivekra1 🐙 github.com/vcodes-git

Education

University of Alberta

Edmonton, Alberta

Bachelor of Science in Computing Science

June 2026

- **Academic Performance:** 3.9 GPA. (First Class Honors, Dean's Silver Medal in Science.)
- **Relevant Coursework:** Algorithms I, Intro to Software Engineering, Network Security, Natural Language Processing, Basics of Machine Learning, File & Database Management.

Projects

3D Architectural Reconstruction Pipeline | *Python, COLMAP, Computer Vision, NumPy* April 2026 | [GitHub](#)

- Eliminated fragmented point cloud failures by replacing sequential brute-force matching with a temporal window-based approach and KD-tree indexed ANN search, reducing matching complexity from $O(N^2)$ to $O(N \log N)$.
- Wrote a script to inject verified matches into COLMAP's SQLite database, correcting radial distortion via OpenCV-to-PINHOLE conversion and applying SOR and DBSCAN outlier removal to achieve sub-pixel reprojection accuracy (0.421px).
- Implemented 3D Gaussian Splatting over 30,000 iterations initialized from the cleaned sparse point cloud, achieving under 2.3% geometric error on CSC window dimensions verified against real-world measurements.

End-to-End NLP & Language Generation Pipeline | *Python, N-Grams, CFG, Regex* Sept 2025 – Dec 2025

- Wrote the core preprocessing pipeline for a phonetic transcription system converting CHILDES transcripts into ARPAbet pronunciations, including a fallback approximator that stitches known sub-word pronunciations for out-of-dictionary words.
- Ported and extended a hand-built context-free grammar and CYK-style parser to classify sentence grammaticality, contributing most of the analysis behind 33.59% precision and 62.59% recall across 603 test cases.
- Co-authored the corpus preprocessing and n-gram training/evaluation modules with two teammates, implementing CLI validation and JSON model serialization and resolving merge conflicts to integrate parallel work.
- Owned the largest share of the language modeling and generation codebase, implementing OOV/UNK thresholding and all three text-generation modes, and tuning interpolation weights via grid search and EM.

Aethon Events Management Platform | *Java, Android Studio, Firebase, NoSQL* Dec 2024 | [GitHub](#)

- Built the app's device-recognition entry point, querying Firestore by device ID to route users to the Admin interface, main app, or sign-up flow based on account role.
- Owned the notification system end-to-end, rewriting an initial MVC scaffold into Firestore-driven logic that builds and delivers per-event notifications, backed by a dedicated JUnit test suite covering notification, profile, and sign-up flows.

GenAI RAG API Microservice | *Python, LangChain, FastAPI, Docker, Azure* Jan 2026 – Apr 2026 | [GitHub](#)

- Built an agentic LLM orchestration pipeline using LangChain and ChromaDB for retrieval-augmented generation, extracting structured business insights from a 10,000-row dataset via local HuggingFace transformer embeddings, and containerized the service with Docker and GitHub Actions CI/CD.
- Deployed the service on Azure Container Apps with Key Vault-backed secret management and a system-assigned managed identity, scoping AcrPull and Key Vault Secrets User roles to enforce least-privilege access at runtime.
- Accelerated development with Claude Code for boilerplate generation and debugging across the FastAPI service and Pydantic schemas.

Experience

Undergraduate Research Assistant

May 2025 – August 2025

University of Alberta (Supervised by Prof. Rupam Mahmood)

Edmonton, Alberta

- Contributed to applied AI research by deploying a streaming actor-critic reinforcement learning policy onto a resource-constrained ESP32-S3 microcontroller, translating a research-stage algorithm into a working real-time system.
- Developed deep working knowledge of an in-house C++ autograd backend and on-device training loop, running experiments and parameter tuning to optimize performance under real hardware constraints.
- Processed high-frequency real-time sensor data and implemented online running statistics for observation and reward normalization, maintaining stable model performance under dynamic operating conditions.

Technical Skills

Languages: Python, Java, C/C++, SQL

AI/ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, LangChain, HuggingFace, NumPy, Pandas, OpenCV

Systems & Security: Linux/Unix, Docker, CI/CD (GitHub Actions), Git, Network Security fundamentals, pwntools

Data & Cloud: Vector Databases (ChromaDB), SQLite/MySQL, Firebase Firestore, Azure (Container Apps, Key Vault)

Practices: Data Structures & Algorithms, Data Preprocessing, Model Evaluation, Agile/Scrum, AI-Assisted Development (Claude Code)